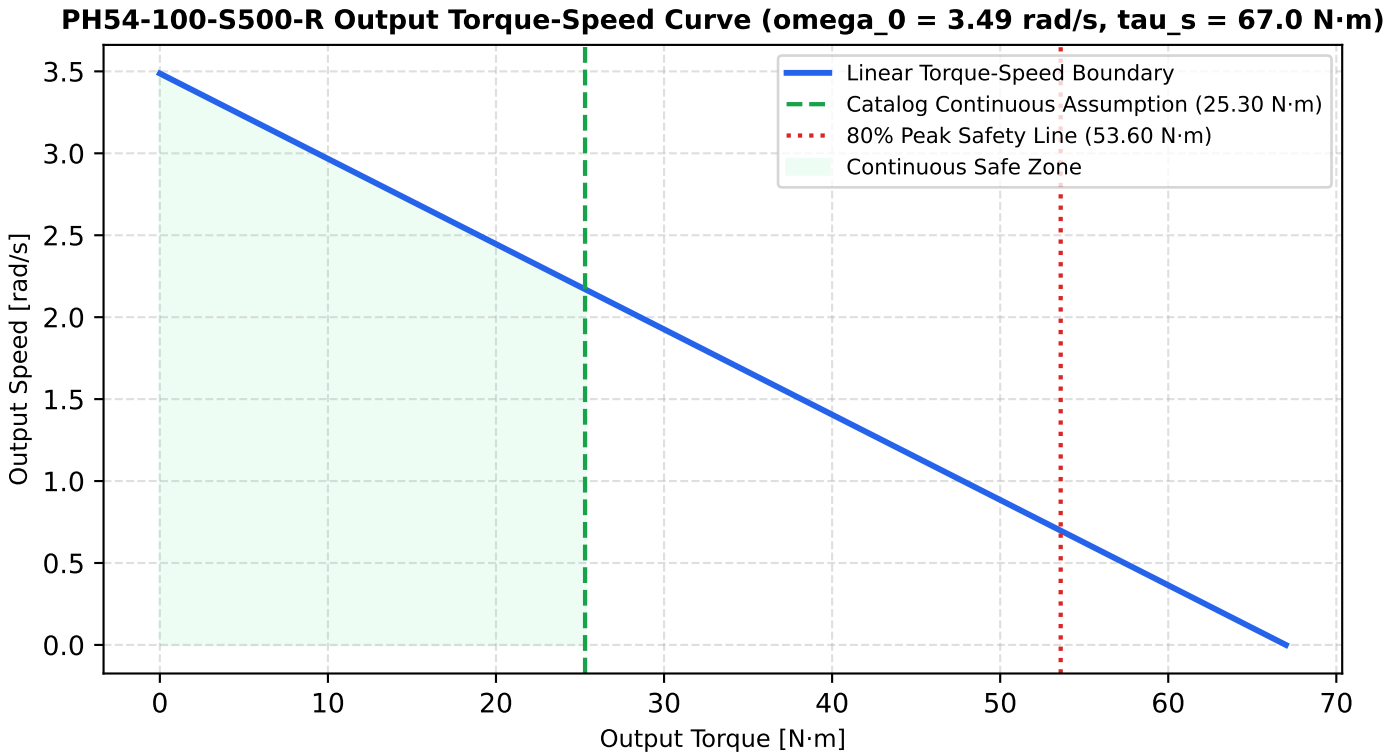
Part Number: PH54-100-S500-R

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	PH54-100-S500-R	DYNAMIXEL-P
Manufacturer	Vendor	ROBOTIS	BLDC Servo
Rated Input Voltage	V_rated	24.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	67.0 N.m peak/stall	At 24.0 V
Continuous Rated Torque	tau_cont	25.3 N.m rated continuous	25 degC ambient, continuous duty
No-Load Speed	omega_0	33.3 rpm (at 24.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	740 g	Includes housing & internal gearhead
Internal Reduction	i_int	502.0 : 1 Cycloidal	Integrated gear stage
Envelope Dimensions	W x H x D	54.0 x 108.0 x 54.0 mm	Outer housing envelope



2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 54.0 mm x 108.0 mm x 54.0 mm
   - Catalog Mass (unverified): 740 g (0.7400 kg)
   - Mounting Pattern: Not verified; obtain manufacturer flange/shaft drawings.

3. Thermal & Duty-Cycle Notes
   - Catalog continuous torque is an input assumption until its manufacturer rating is check
   - The 0.33x stall estimate, where used, is a heuristic rather than a measured thermal rat
   - The linear torque-speed envelope is a screening model, not a measured performance curve

4. Project Reference Index
   - Part ID: PH54-100-S500-R | Source Document: ph54-100-s500-r.pdf | Spec Table: Page 1